

Block-Sparse Attention with Prefix Caching: Integrating MInference into vLLM via the PagedAttention–Sparsity Isomorphism

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I. Purpose of the Project

Large Language Models increasingly demand long-context reasoning (128K–10M tokens). In production, **shared system prompts** are identical across requests, yet must be reprocessed every time—a fundamental bottleneck.

Two complementary optimisations exist independently:

- **PagedAttention** (vLLM): Stores KV caches in fixed-size blocks; *prefix caching* computes shared KV once and reuses it across requests.
- **MInference**: Exploits the sparsity of attention weights (<5% non-zero), reducing FLOPs by up to 95% via dynamic block-sparse attention.

Problem. These systems are *fundamentally incompatible*. MInference requires equal-length Q, K, V processed from scratch—no mechanism to accept a pre-computed KV cache.

Objective. Identify the structural isomorphism between block-sparse attention and PagedAttention, enabling sparse attention + prefix caching for the first time.

II. The PagedAttention–Sparsity Isomorphism

Block-sparse attention is **naturally compatible** with PagedAttention’s paged memory layout:

- Block-sparse attention partitions the attention matrix into blocks of size $b \times b$; column c spans KV tokens $[cb, (c+1)b)$.
- PagedAttention stores KV vectors in physical pages of B tokens; page $\phi(c)$ stores tokens $[cB, (c+1)B)$.

Setting $b = B$, the correspondence is exact:

Attention Block $(r, c) \longleftrightarrow$ Physical KV Page $\phi(c)$

No data reorganisation, no gather operation, no auxiliary structures. The block table ϕ provides $O(1)$ access to the exact physical memory for any block. Selective page access reduces I/O dramatically:

$$\mathcal{P}_{\text{load}} = \{ \phi(c) : \exists r \text{ s.t. } (r, c) \in \hat{\mathcal{I}} \} \quad |\mathcal{P}_{\text{load}}| \leq k_b \cdot \frac{n_q}{B} \ll \frac{n_p + n_q}{B} \cdot \frac{n_q}{B}$$

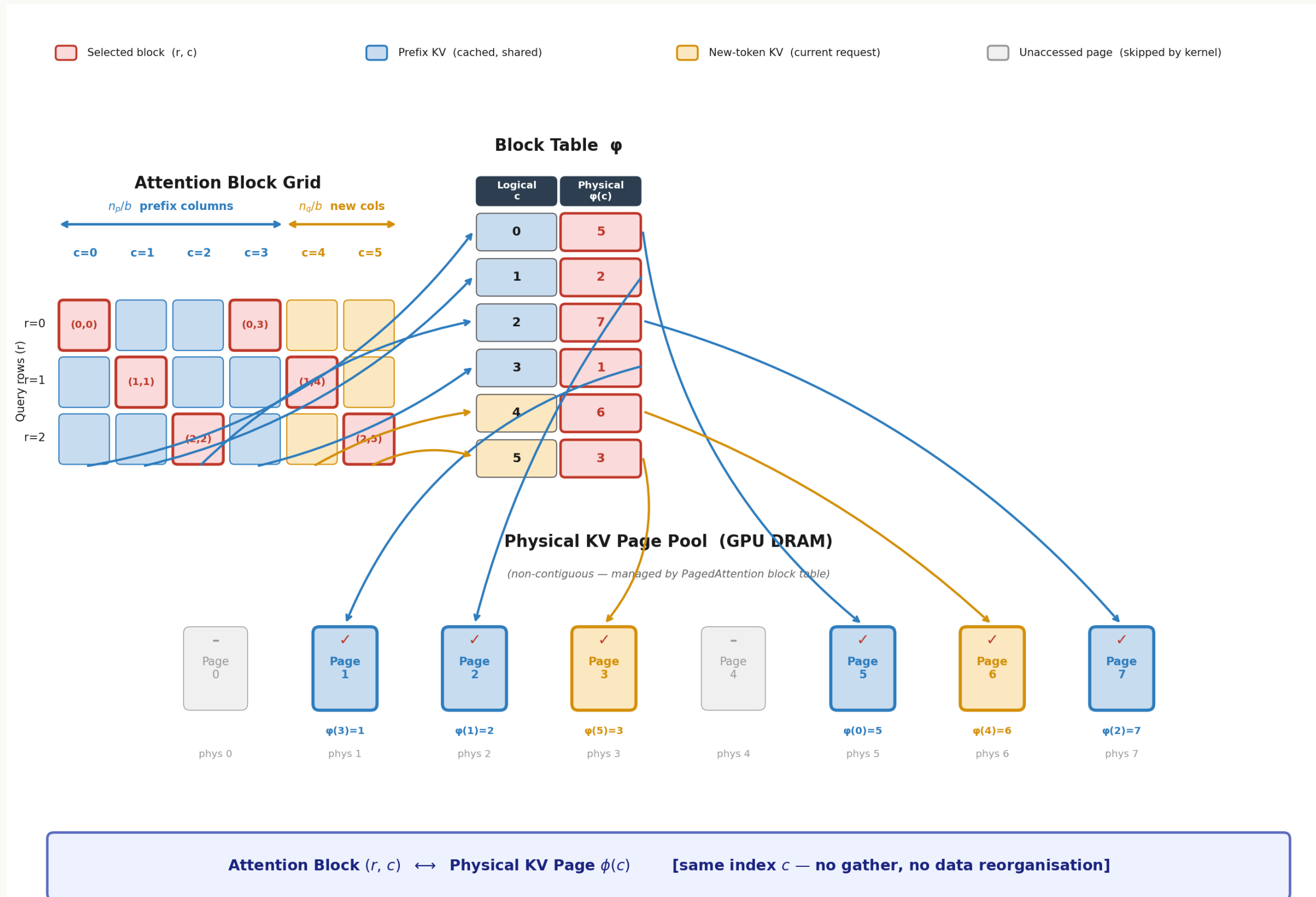


Figure 1. KV vectors for block (r, c) are precisely stored in physical page $\phi(c)$.

Prefix caching and shared pages. Shared prefix pages are stored once in physical memory and referenced by multiple requests via the block table. Each request’s importance predictor independently selects a different subset of prefix pages depending on its query content. Pages not selected are simply not accessed—with zero overhead.

Contributions

1. Identify the fundamental incompatibility between MInference and KV cache systems.
2. Formalise the structural isomorphism between block-sparse column indices and PagedAttention page addresses.
3. Integrate MInference’s predictor into vLLM, enabling sparse attention over prefix-cached KV pages for the first time.
4. Demonstrate 63% latency reduction with near-identical output fidelity.

III. Methodology

The system integrates into vLLM with three components:

1 Online Importance Predictor

Coarse block importance via mean-pooled Q and K over the full rectangular grid:

$$\bar{Q}_r = \frac{1}{b} \sum_{i=rb}^{(r+1)b} Q_i, \quad \bar{K}_c = \frac{1}{b} \sum_{j=cb}^{(c+1)b} K_j, \quad \hat{A}_{rc} = \frac{\bar{Q}_r \bar{K}_c^\top}{\sqrt{d_h}}$$

Top- k_b columns per query row yield $\hat{\mathcal{I}}$.

2 Sparse Page Selector

Translates $\hat{\mathcal{I}}$ into physical pages via direct block-table lookup:

$$\mathcal{C} = \{ c : \exists r \text{ s.t. } (r, c) \in \hat{\mathcal{I}} \} \xrightarrow{\phi} \{ \phi(c) : c \in \mathcal{C} \}$$

At 5% sparsity $\Rightarrow 20\times$ reduction in pages accessed.

3 Block-Sparse FlashAttention Kernel

Online softmax over selected blocks only; unselected blocks incur **zero computation**:

$$O_r \approx \frac{\sum_{c \in \hat{\mathcal{I}}} e^{S_{rc} - m_r} V_c}{\sum_{c \in \hat{\mathcal{I}}} e^{S_{rc} - m_r}}$$

IV. Findings

Theoretical Speedup.

$$\frac{t_{\text{proposed}}}{t_{\text{MInference}}} = \frac{n_q}{n_p + n_q} \quad \text{As prefix grows, latency} \rightarrow 0.$$

Empirical Results (NVIDIA RTX 3090, bf16):

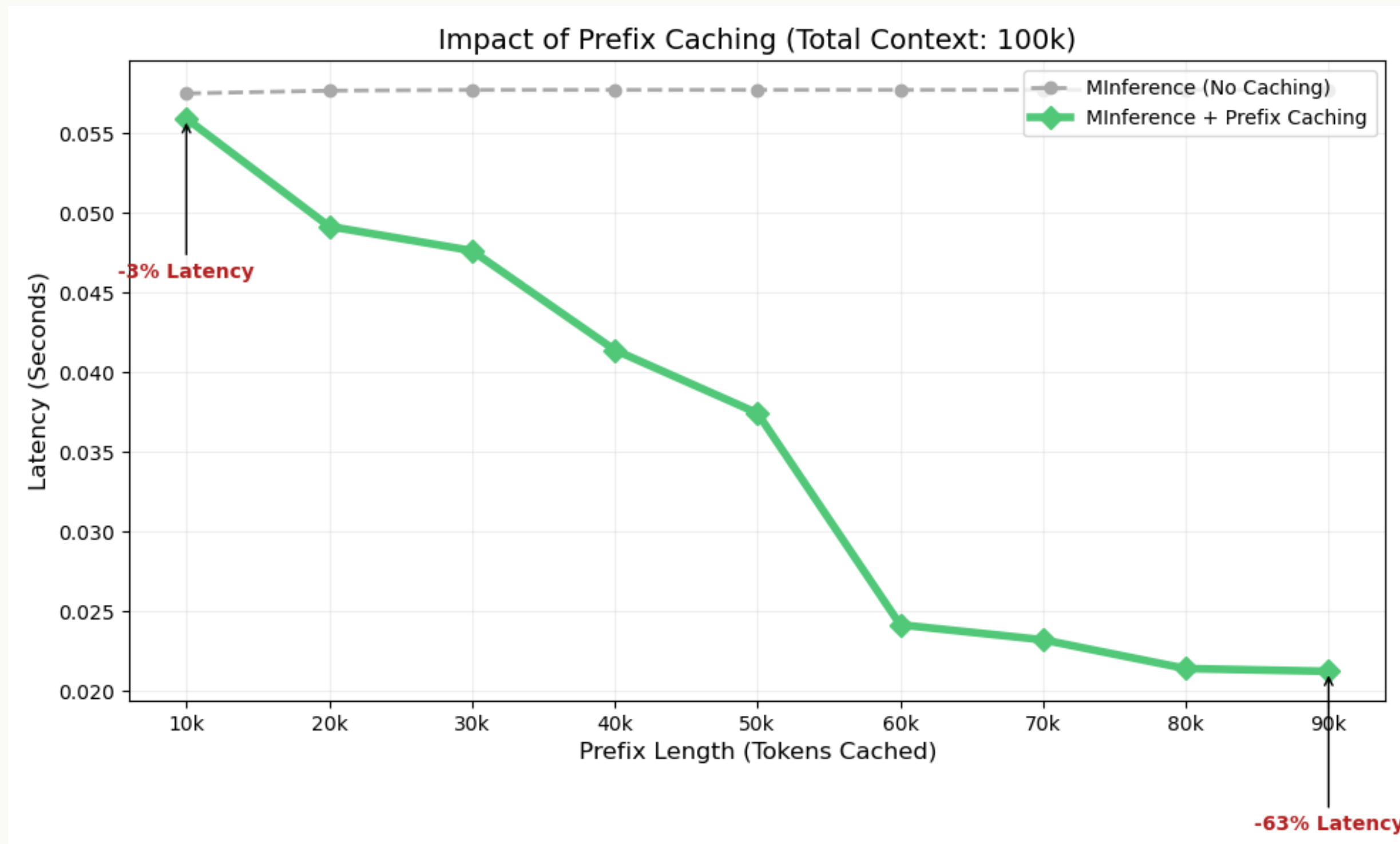


Figure 2. Latency at 100K context.

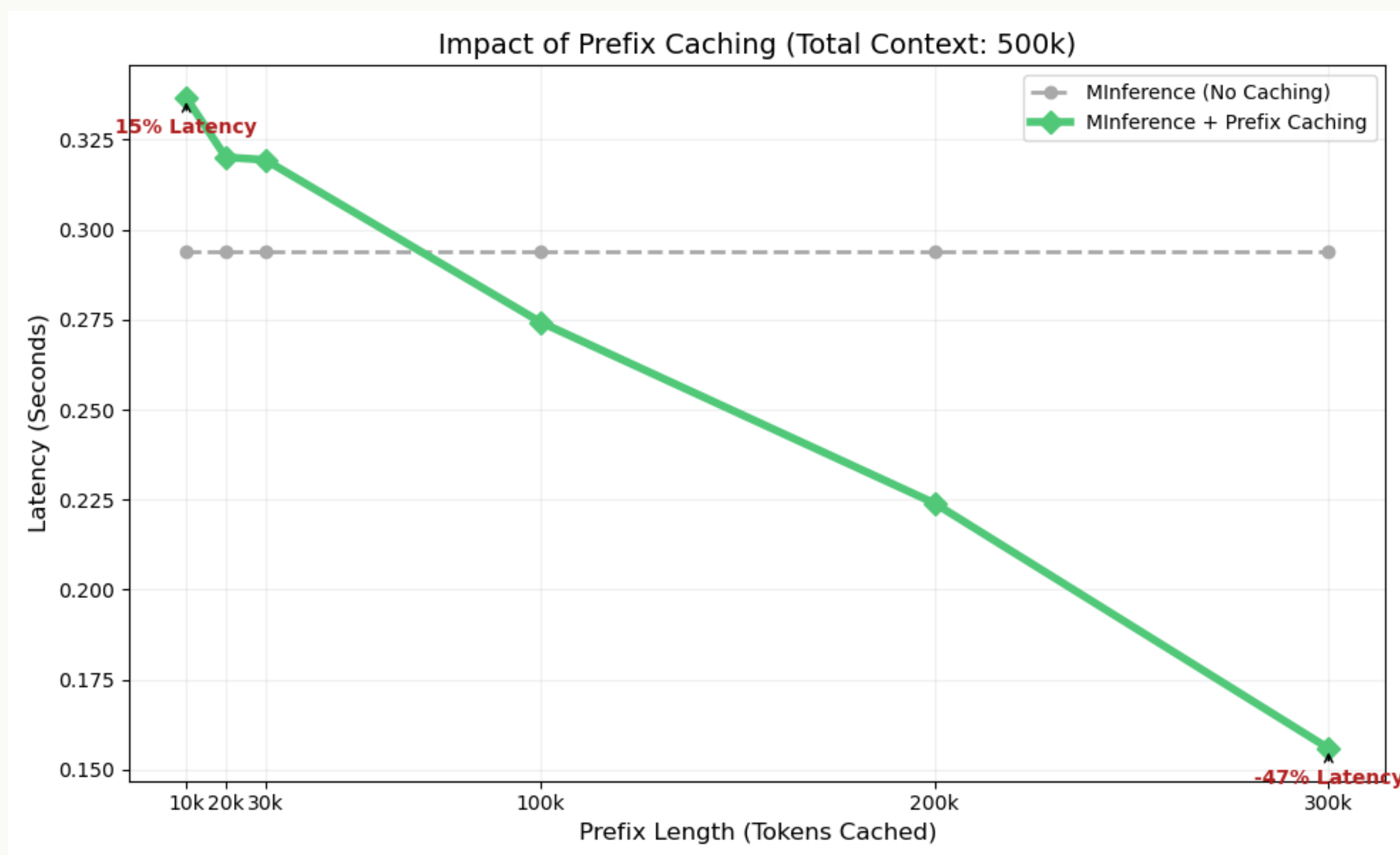


Figure 3. Latency at 500K context.

Key Results

- **63% reduction** in attention kernel latency (100K tokens, 90K prefix cache).
- Near-identical output fidelity: $\max |\Delta| \approx 2 \times 10^{-3}$ (bf16).
- Speedup scales **monotonically** with prefix length.
- Figures are a *lower bound*—end-to-end speedup is larger (MLP savings excluded).

Conclusion

This work identifies and formalises the **PagedAttention–block-sparsity isomorphism**: the column index of an attention block corresponds directly to the physical KV page $\phi(c)$ in PagedAttention’s block table. This natural alignment enables block-sparse attention to operate directly over paged, prefix-cached KV memory—without data reorganisation or auxiliary structures. By integrating MInference’s importance predictor into vLLM, the system selectively attends only to the most important pages per query row, achieving up to **63% latency reduction** while maintaining near-identical output fidelity. This isomorphism constitutes a general **design principle**: any block-structured KV memory management system can directly leverage block-level importance scores to guide selective attention computation.

Limitations & Future Work

The primary limitation is the overhead of online importance prediction, which grows as $O((n/b)^2 d_h)$ and may become non-trivial for extremely long contexts ($n > 1\text{M}$). Future work includes hierarchical prediction schemes, learned static masks per head, and formal end-to-end evaluation on long-context benchmarks (RULER, InfiniteBench).